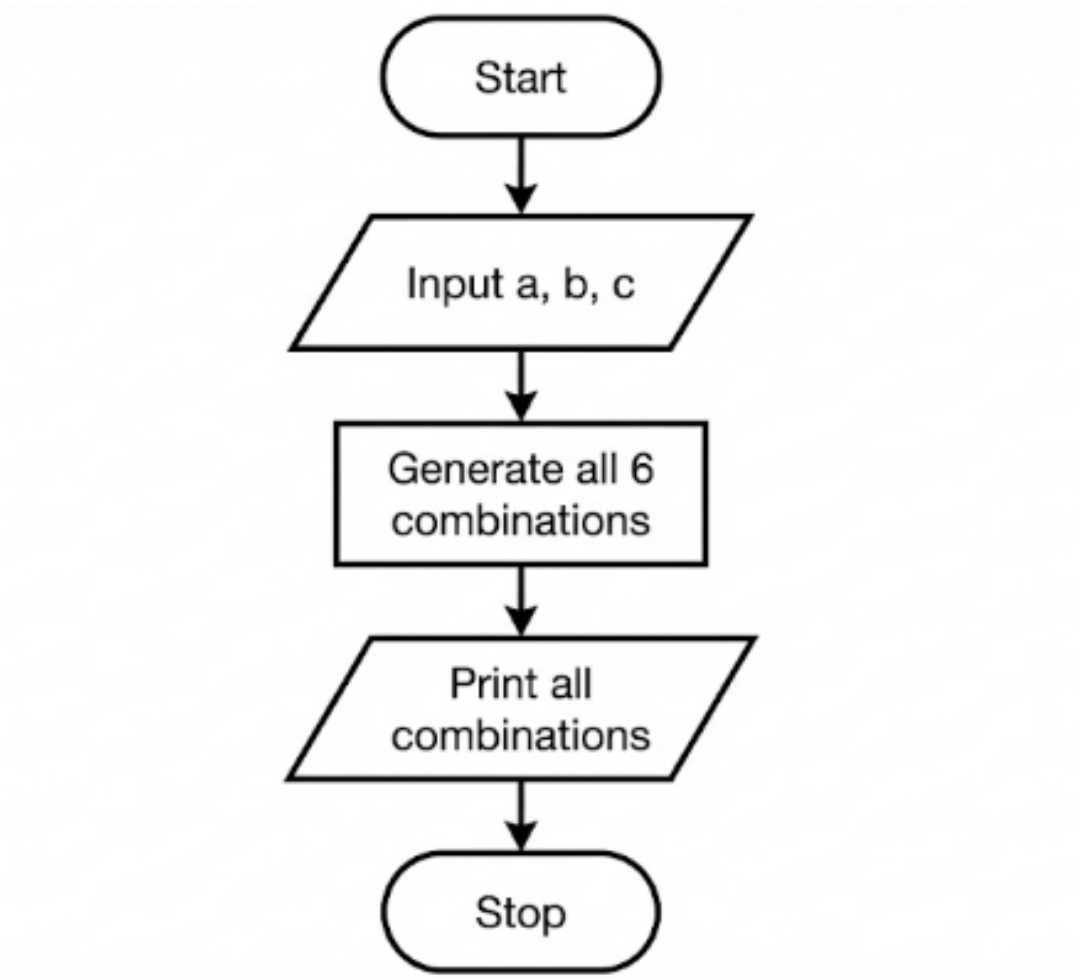
Experiment 7.2: Possible combination from 3 digits

Algorithm:

```
Step 1: START
Step 2: INPUT a, b, c  (three space-separated strings)
Step 3: STORE  digits = [a, b, c]
Step 4: FOR i = 0 TO 2:
            FOR j = 0 TO 2:
                FOR k = 0 TO 2:
                    IF i != j AND j != k AND i != k:
                        PRINT  digits[i] + digits[j] + digits[k]
Step 5: STOP
```

Flowchart:



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7.1.2. Possible Combinations from Three Digits

02:08

Write a Python program to accept three digits and print all possible 3-digit combinations that can be formed using those three digits. Each digit should be used exactly once in each combination.

For three digits a, b, and c, there are 6 possible arrangements based on their indices: abc, acb, bac, bca, cab, cba.

Input Format:

- Single line contains three space-separated non-negative integers, a, b, and c, representing the three digits

Output Format:

- Print all possible 3-digit combinations
- Each combination should be printed as a 3-digit number on a separate line
- Print combinations as they appear (maintain leading zeros if first digit is 0)

Constraints:

- $0 \leq \text{each digit} \leq 9$
- Digits can be same or different

Sample Test Cases +

Explorer

combinat...

Submit

```python
a, b, c = input().split()

digits = [a, b, c]

for i in range(3):
    for j in range(3):
        for k in range(3):
            if i != j and j != k and i != k:
                print(digits[i] + digits[j] + digits[k])
```

Debugger

Average time	Maximum time
0.014 s	0.017 s
14.14 ms	17.00 ms

3 out of 3 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 17 ms Debug

Expected output	Actual output
1 2 3	1 2 3
123	123
132	132
213	213
231	231
312	312

Terminal Test cases

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Code: